

The Config Bench

Lab 3 — Series & Parallel Circuits | On-Camera Script | Rev 4

Episode TCB-LAB-003 | John King | @TheConfigBench | Target: <10 min

Rev 4 Changes: Qucs DC operating point simulation added at end of each schematic segment (results annotated directly on schematic on PM1610). One breadboard at a time — no side-by-side reference. Tolerance teaching point added to each live breadboard demo. Ohm's Law next-episode tease added throughout and in close. Giveaway updated — oscilloscope prize, winner announced soon, test questions in the Ohm's Law episode.

Episode Timing Overview

Segment	Content	Est. Time	Running Total
1	Welcome + bench pan	0:30	0:30
2	Lab 1 schematic + Qucs DC sim	1:15	1:45
3	Lab 2 schematic + Qucs DC sim	1:00	2:45
4	Lab 3 schematic + Qucs DC sim	1:15	4:00
5	Breadboard 1 live demo + tolerance note	1:00	5:00
6	Breadboard 2 live demo + tolerance note	1:00	6:00
7	Breadboard 3 live demo + tolerance note	1:00	7:00
8	DSO152 oscilloscope quick demo	0:45	7:45
9	Close — Ohm's Law tease, giveaway, subscribe	1:00	8:45
	Buffer / edit room	1:15	~10:00

Pre-Production Checklist — TCB-LAB-003

Print this page and keep it at your bench. Work top to bottom. Day-Before items done the night prior so shoot day starts clean.

DAY BEFORE — Evening Prep

[]	Charge Canon EOS 60D battery <i>Fully charged LP-E6 in camera body. Spare charged if available.</i>
[]	Charge iPhone 14 Pro <i>100% charge. Plug in overnight. Consider keeping plugged in during shoot.</i>
[]	Charge Flyfrost 8K 88MP UHD battery <i>Fully seated in charger overnight. Confirm power-on before shoot.</i>
[]	Charge FNIRSI DSO152 oscilloscope <i>USB charge. Confirm power-on and probe connection before shoot day.</i>
[]	Confirm memory cards are formatted and installed <i>Canon EOS 60D: SD card seated and formatted. Flyfrost 8K: storage confirmed. iPhone: sufficient free space.</i>
[]	Pre-build all three breadboards <i>Board 1: single 47Ω. Board 2: two 47Ω series. Board 3: four 470Ω (2S2P). Do NOT connect batteries yet.</i>
[]	Load all three Qucs schematics and run DC sim on each <i>Confirm results annotate correctly on each schematic. Save each sim result. Know which file to open for each segment.</i>
[]	Load schematics on Acer PM1610 <i>Three Qucs files ready to switch: Lab 1, Lab 2, Lab 3. Test display brightness and cable connection.</i>
[]	Check 9V battery voltage pre-load <i>Measure with Sperry DM-210A. Note the reading — you will use this number on camera.</i>
[]	Confirm ANENG SZ308 and Sperry DM-210A are working <i>Fresh batteries in both meters if needed. Test on a known source.</i>

SHOOT DAY — Room & Bench Setup

[]	Set up camera arm positions <i>Use the marker positions on your bench. Left arm: iPhone 14 Pro. Right-center arm: Flyfrost 8K. Far-right arm: Canon EOS 60D.</i>
[]	Mount all three cameras on arms <i>Secure each camera head. Check arm tightness — no drift during shoot.</i>
[]	Set bench lighting <i>Even, shadow-free illumination across bench surface. No glare on meter displays or PM1610 monitor.</i>
[]	Position Acer PM1610 <i>Behind equipment, visible in background on Canon wide shot. Load Lab 1 Qucs schematic (pre-sim) to start.</i>

[]	Place breadboard on angled stand <i>One board at a time — swap during shoot. Start with Board 1 on stand.</i>
[]	Position Sperry DM-210A and ANENG SZ308 <i>Both meters in frame on Canon wide shot. Probes ready — NOT yet connected.</i>
[]	Position FNIRSI DSO152 <i>To the RIGHT of the breadboard stand. Probe attached. Powered on. Visible in frame but not featured until Segment 8.</i>
[]	Place pencil/pointer within reach <i>Used for schematic walk on PM1610. Keep it next to the monitor.</i>

SHOOT DAY — Camera Checks (Do In Order)

[]	Canon EOS 60D — angle and framing <i>Overhead wide. Full bench in frame: breadboard on stand, both meters, DSO152, PM1610 visible in background.</i>
[]	Canon EOS 60D — settings <i>1920x1080, 30fps, 1/60s shutter, manual white balance matched to bench lighting.</i>
[]	Canon EOS 60D — memory card confirmed <i>SD card inserted, formatted, sufficient space for a 12-minute buffer.</i>
[]	Flyfrost 8K — angle and framing <i>Overhead wide right-center. Full lab surface visible. Confirm focus and exposure.</i>
[]	Flyfrost 8K — memory card confirmed <i>Card inserted, formatted, sufficient space.</i>
[]	iPhone 14 Pro — angle and framing <i>Left arm, tight close-up. Confirm it can reach: breadboard connections, meter probes, PM1610 screen, DSO152 screen.</i>
[]	iPhone 14 Pro — storage and airplane mode <i>Sufficient free space. Airplane mode ON to prevent notification interruptions.</i>
[]	Test record all three cameras — 30 seconds <i>Play back. Confirm angle, focus, and exposure on each before going live.</i>

SHOOT DAY — Audio

[]	Microphone connected <i>Confirm mic cable is seated — camera body or audio interface.</i>
[]	Microphone powered on <i>Battery or phantom power confirmed ON. LED indicator check.</i>
[]	10-second audio test record <i>Play back — confirm clean audio, no hum, no clipping.</i>
[]	Room ambient noise check <i>HVAC off or noted. No phone notifications. Bench area quiet before rolling.</i>

FINAL GO / NO-GO — Last Check Before Rolling

☐	All three cameras recording
☐	Audio confirmed hot
☐	PM1610 showing Lab 1 Qucs schematic — pre-simulation view
☐	Board 1 on stand — battery NOT yet connected
☐	Boards 2 and 3 staged nearby — ready to swap in
☐	Sperry and ANENG in frame, probes ready
☐	DSO152 powered on, probe attached, in frame
☐	Pencil within reach of PM1610
☐	9V battery voltage noted — 8.2V pre-load confirmed
☐	You are ready. Hit record.

Production Setup — Camera & Gear Reference

Camera / Item	Details
Cam 1 — iPhone 14 Pro	LEFT arm — tight close-up: breadboard connections, meter probes, PM1610 screen, DSO152 screen
Cam 2 — Flyfrost 8K	RIGHT CENTER arm — overhead wide shot of full lab surface
Cam 3 — Canon EOS 60D	FAR RIGHT arm — overhead cinematic wide — primary edit cut
Acer PM1610	Background monitor — Qucs schematic display. Switch between Lab 1, 2, 3 files as each segment begins. Run DC sim live on camera.
Breadboard 1 (pre-built)	Single 47Ω resistor — on angled stand, wired, NOT connected to battery at start
Breadboard 2 (pre-built)	Two 47Ω resistors in series — staged nearby, swapped in after Board 1 demo
Breadboard 3 (pre-built)	4x 470Ω (2S2P parallel) — staged nearby, swapped in after Board 2 demo
Sperry DM-210A	Voltage measurement only — probes across circuit terminals
ANENG SZ308	Current measurement — wired in series before powering on
9V Battery	Pre-load = 8.2V — show this reading on camera before first board
FNIRSI DSO152	To RIGHT of breadboard stand — powered on, probe attached, in frame. Not featured until Segment 8.
Pencil / pointer	Used throughout schematic walk and Qucs sim callouts on PM1610

Segment 1 — Welcome & Bench Pan

■ Segment 1 | Est. ~0:30

■ Canon EOS 60D wide overhead. Bench dressed — Board 1 on angled stand, both meters in frame, DSO152 to right of stand, PM1610 glowing in background.

JOHN: Welcome to The Config Bench. I'm John — and this is Lab 3.

JOHN: If you're just joining us — Labs 1 and 2 are linked in the description. Go watch those first. This lab builds directly on them.

■ Slow pan across the bench — breadboard on stand, meters, DSO152, PM1610 in background.

JOHN: Today we're going to look at series and parallel circuits using three pre-built breadboards, two inexpensive meters, and a nine volt battery. Everything you see on this bench came in under sixty dollars. Let's get into it.

Segment 2 — Lab 1 Schematic Review + Qucs DC Simulation

■ Segment 2 | Est. ~1:15

■ Pan to PM1610. Lab 1 Qucs schematic on screen — battery and single 47Ω resistor, simulation results NOT yet shown. iPhone close-up on monitor. Pencil ready.

JOHN: Quick review. Lab 1 — the simplest circuit possible. A battery and a single forty-seven ohm resistor.

■ Pencil traces the circuit path — battery positive, through the resistor, back to negative.

JOHN: Current flows from positive, through the resistor, back to negative. Ohm's Law — voltage equals current times resistance.

■ Pencil points to voltage meter placement on schematic.

JOHN: For voltage — the Sperry probes go here, across the resistor. For current — the ANENG goes in series, right here, so all current flows through it.

■ QUCS: Run DC operating point simulation. Results annotate directly onto the schematic on PM1610 — node voltages and branch current visible on screen.

■ iPhone close-up on PM1610 showing annotated schematic with Qucs results. Pencil points to each annotated value.

JOHN: Now — this is where Qucs earns its place on the bench. I'm going to run the DC operating point simulation right here.

JOHN: Watch the schematic. The software is now showing us the calculated voltage at each node and the current through the circuit — automatically. No manual math required.

■ Pencil points to the node voltage annotation on the schematic.

JOHN: Voltage here — [READ QUCS VALUE]. And current through the resistor — [READ QUCS VALUE]. That is what Ohm's Law predicts for this circuit at nine volts. Now let's see what the real bench says. Coming up in a moment.

Segment 3 — Lab 2 Schematic Review + Qucs DC Simulation

■ Segment 3 | Est. ~1:00

■ Switch PM1610 to Lab 2 Qucs schematic — battery and two 47Ω resistors in series. Simulation results NOT yet shown. Pencil ready.

JOHN: Lab 2 — two forty-seven ohm resistors in series. Same battery, but now resistance doubles and current drops in half.

■ Pencil traces series path through both resistors.

JOHN: In series — current is the same through every component. Voltage splits — half across R1, half across R2.

■ Pencil taps voltage meter placement across each resistor and across the full circuit. Then current meter placement in series.

JOHN: Voltage meter here for R1. Here for R2. Here for the full circuit. Current meter in series — same point as before.

■ QUCS: Run DC operating point simulation on Lab 2 schematic. Qucs annotates node voltages across R1, R2, and full circuit, plus branch current, directly onto the schematic.

■ iPhone close-up on PM1610 showing annotated Lab 2 schematic. Pencil points to each value.

JOHN: Let's run the simulation. Qucs is showing us the node voltage here at R1 — [READ QUCS VALUE]. Here at R2 — [READ QUCS VALUE]. And the current through the circuit — [READ QUCS VALUE].

JOHN: Notice the voltage split. Each resistor carries exactly half. That is the series circuit behaving exactly as the theory predicts. Let's keep going.

Segment 4 — Lab 3 Schematic Introduction + Qucs DC Simulation

■ Segment 4 | Est. ~1:15

■ Switch PM1610 to Lab 3 Qucs schematic — battery and four 470Ω resistors in 2S2P configuration. Simulation results NOT yet shown. iPhone close on monitor. Pencil ready.

JOHN: Lab 3 — this is where it gets interesting. Four resistors — four hundred and seventy ohms each. But they're not all in series.

■ Pencil traces Branch 1 — left pair of resistors in series.

JOHN: Branch one — two resistors in series. Four-seventy plus four-seventy — nine hundred and forty ohms on this path.

■ Pencil traces Branch 2 — right pair in series.

JOHN: Branch two — same thing. Another nine hundred and forty ohms. Both branches connect at the same two points — that makes them parallel.

■ Pencil traces both parallel connection points.

JOHN: In parallel — voltage is the same across every branch. Current splits. Total resistance — nine forty divided by two — four hundred and seventy ohms.

■ Pencil points to voltage meter placement across the full circuit, then current meter in series at the supply.

JOHN: Voltage meter across the full circuit. Current meter in series at the supply — it sees both branches combined.

■ QUCS: Run DC operating point simulation on Lab 3 schematic. Qucs annotates node voltages across each branch and at each node, plus total supply current and individual branch currents, directly on the schematic.

■ iPhone close-up on PM1610 showing annotated Lab 3 schematic. Pencil traces each annotated value.

JOHN: Let's see what Qucs calculates for this one. Voltage across Branch 1 — [READ QUCS VALUE]. Branch 2 — [READ QUCS VALUE]. Same — exactly as the theory says.

JOHN: Current through Branch 1 — [READ QUCS VALUE]. Branch 2 — [READ QUCS VALUE]. Total current from the supply — [READ QUCS VALUE]. Both branches sharing the load equally.

JOHN: That's the math. Now let's go see what the real bench does. Alright — over to the breadboards.

Segment 5 — Breadboard 1 Live Demo (Single 47Ω)

■ Segment 5 | Est. ~1:00

■ Pan from PM1610 back to bench. Canon wide. PM1610 stays on Lab 1 schematic with Qucs results visible in background. Board 1 on angled stand. Sperry and ANENG in frame.

JOHN: Breadboard 1. Single forty-seven ohm resistor. Pre-built — and the schematic with Qucs calculations is right there behind me for reference.

■ iPhone close-up on 9V battery. John connects battery leads to Board 1.

JOHN: Battery reads eight point two volts pre-load. Not nine. Batteries sag under use. Always measure first.

■ *Connect battery. iPhone close-up on Sperry DM-210A. Let reading settle. Read aloud.*

JOHN: Power on. Voltage across the resistor — [READ METER ALOUD].

■ *iPhone close-up on ANENG SZ308. Read current aloud.*

JOHN: Current — [READ METER ALOUD].

■■ **Tolerance Teaching Point:** The voltages and currents you see here will adhere to Ohm's Law but may differ slightly from what Qucs calculated. These variations are caused by component tolerances in the resistors, and the tolerance of the components inside the meters themselves. Do not get hung up on slight differences — this is normal, real-world electronics. Our next video will go deeper into Ohm's Law and this will all start to make sense.

JOHN: You may notice those numbers are close to what Qucs predicted but not identical. That's completely normal. Your resistors have a tolerance — typically five percent. And the meters themselves have tolerances built into their components. Don't get hung up on small differences. The values adhere to Ohm's Law — and our next video is going to go deeper into exactly why this happens.

Segment 6 — Breadboard 2 Live Demo (Two 47Ω Series)

■ **Segment 6 | Est. ~1:00**

■ *Disconnect Board 1 battery. Remove Board 1 from stand. Place Board 2 on stand. PM1610 switches to Lab 2 schematic with Qucs results. Canon wide.*

JOHN: Breadboard 2. Two forty-seven ohm resistors in series. Swapping the board — same stand, same camera angle. Lab 2 schematic with the Qucs results is up behind me.

■ *Connect battery to Board 2. iPhone close-up on Sperry.*

JOHN: Voltage across the full circuit — [READ METER ALOUD]. Same source voltage.

■ *Move Sperry probes to R1 only. iPhone close-up.*

JOHN: Across R1 alone — [READ METER ALOUD]. Across R2 — [READ METER ALOUD]. Add them together and you're back to your source. Series — voltage divides, current stays the same.

■ *iPhone close-up on ANENG SZ308.*

JOHN: Current — [READ METER ALOUD]. About half of Board 1 — because we doubled the resistance.

■■ **Tolerance Teaching Point:** The voltages and currents you see here will adhere to Ohm's Law but may differ slightly from what Qucs calculated. These variations are caused by component tolerances in the resistors, and the tolerance of the components inside the meters themselves. Do not get hung up on slight differences — this is normal, real-world electronics. Our next video will go deeper into Ohm's Law and this will all start to make sense.

JOHN: Again — compare these readings to what Qucs showed us. You'll see they're close but not identical. Component tolerances, meter tolerances — perfectly normal. This is real electronics, not a textbook. Our next video on Ohm's Law will help tie this all together.

Segment 7 — Breadboard 3 Live Demo (4x 470Ω Parallel)

■ Segment 7 | Est. ~1:00

■ *Disconnect Board 2. Remove from stand. Place Board 3 on stand. PM1610 switches to Lab 3 parallel schematic with Qucs results. DSO152 visible to right of stand. Canon wide.*

JOHN: Breadboard 3. Four four-hundred-and-seventy ohm resistors — two in series per branch, two branches in parallel. Lab 3 schematic with Qucs results is right behind me.

■ *iPhone close-up on Board 3. Pencil points out the two parallel branches on the physical board.*

JOHN: Two paths right here. Both see the same voltage. Current splits between them.

■ *Connect battery. iPhone close-up on Sperry.*

JOHN: Power on. Voltage — [READ METER ALOUD]. Same source. Parallel — voltage holds across both branches.

■ *iPhone close-up on ANENG SZ308.*

JOHN: Total current — [READ METER ALOUD]. Both branches combined. Lowest draw of the three boards.

■ ■ **Tolerance Teaching Point:** The voltages and currents you see here will adhere to Ohm's Law but may differ slightly from what Qucs calculated. These variations are caused by component tolerances in the resistors, and the tolerance of the components inside the meters themselves. Do not get hung up on slight differences — this is normal, real-world electronics. Our next video will go deeper into Ohm's Law and this will all start to make sense.

JOHN: Same story as before — your readings will be very close to the Qucs simulation but not a perfect match. Resistor tolerances, meter tolerances. The circuit is behaving exactly according to Ohm's Law. We'll prove that in detail in the next video. Keep that board powered — one more thing to show you.

Segment 8 — FNIRSI DSO152 Oscilloscope Quick Demo

■ Segment 8 | Est. ~0:45

■ *Board 3 still powered. John picks up DSO152. Probe connected across Board 3 rails. iPhone close-up on DSO152 screen.*

JOHN: One more tool. This is the FNIRSI DSO152 — a pocket oscilloscope. Part of that sixty dollar bench.

■ *Hold DSO152 toward Camera 1. Show the flat DC trace on screen.*

JOHN: What you're seeing is the DC voltage of our circuit. That flat line — direct current. Steady. No waveform.

JOHN: A multimeter gives you a number. The oscilloscope shows you the shape of the signal over time. For DC — it's flat. That's it.

JOHN: I'm not going deep on this today — that's its own dedicated lab. But in a future video we're going to use this to look at AC signals and waveforms that a multimeter simply cannot show you. Stay tuned for that one.

Segment 9 — Close

■ Segment 9 | Est. ~1:00

■ Canon EOS 60D wide. Board 3 still on stand. Both meters visible. DSO152 in frame. PM1610 showing TheConfigBench logo or Lab 3 summary.

- JOHN:** Three circuits. Three breadboards. Two meters. A pocket oscilloscope. A nine volt battery. Under sixty dollars.
- JOHN:** Lab 1 — one resistor, maximum current. Lab 2 — series, voltage divides, current drops. Lab 3 — parallel, voltage holds, current splits. Same battery. Three completely different behaviors — all because of how the resistors are connected.
- JOHN:** In the next video we're going to go into Ohm's Law in detail — and that's going to tie everything you saw today together. The Qucs simulations, the meter readings, the tolerances — it's all going to make a lot more sense after that one. That's also when we're going to run our test questions, so pay attention.
- JOHN:** And speaking of the next video — we'll be announcing the winner of this oscilloscope very soon. To get your name in — leave a comment below. Tell me which of these three circuits surprised you the most.
- JOHN:** Subscribe so you don't miss the Ohm's Law video or the giveaway announcement. I'm John. This is The Config Bench. Thanks for watching.

On-Camera Reference — Expected Measurements at 8.2V

Calculated using 8.2V pre-load. Read actual meter values aloud — any difference vs Qucs is the tolerance teaching point. Use Qucs 9V sim values as the theoretical baseline on camera.

Board	Circuit	Total R	Expected Voltage	Expected Current	Notes
1	Single 47Ω	47Ω	~8.2V across R1	~174 mA	Highest current — R may feel warm
2	Two 47Ω series	94Ω	~4.1V each (R1 and R2)	~87 mA	Show voltage split across each R individually
3	4x 470Ω (2S2P)	470Ω	~8.2V across each branch	~17.4 mA total ~8.7 mA per branch	Lowest current — two parallel paths

YouTube Metadata

Field	Content
Title Option 1	Series vs Parallel — Real Circuits on a \$60 Bench Lab 3 — The Config Bench
Title Option 2	Ohm's Law in Action: 3 Breadboards, 2 Meters, \$60 Total TCB Lab 3
Title Option 3	Lab 3: Parallel Circuits + Qucs Simulation The Config Bench
Description	In Lab 3 we compare series and parallel circuits using three pre-built breadboards, Qucs DC simulations, a Sperry DM-210A, ANENG SZ308, and FNIRSI DSO152 — all under \$60. Next up: Ohm's Law deep dive.
Tags	series circuit, parallel circuit, Ohm's Law, breadboard, Qucs simulator, electronics for beginners, ANENG SZ308, Sperry DM-210A, FNIRSI DSO152, 470 ohm, TheConfigBench, budget electronics
End Screen	Link to Lab 1, Lab 2, Subscribe button, tease Ohm's Law next video
Pinned Comment	Giveaway: FNIRSI DSO152 Oscilloscope — comment which circuit surprised you most to enter! Winner announced soon.
Chapter Markers	0:00 Welcome 0:30 Lab 1 Schematic + Qucs 1:45 Lab 2 Schematic + Qucs 2:45 Lab 3 Schematic + Qucs 4:00 Board 1 Live 5:00 Board 2 Live 6:00 Board 3 Live 7:00 Oscilloscope Demo 7:45 Close